DP Biology – D3.2 (HL extension)

Two genes at a time

- What patterns of inheritance are seen in plants and animals?
- What happens at the molecular level to produce these patterns?

HL statements D3.2.16–21. Textbook pp. 747–759.

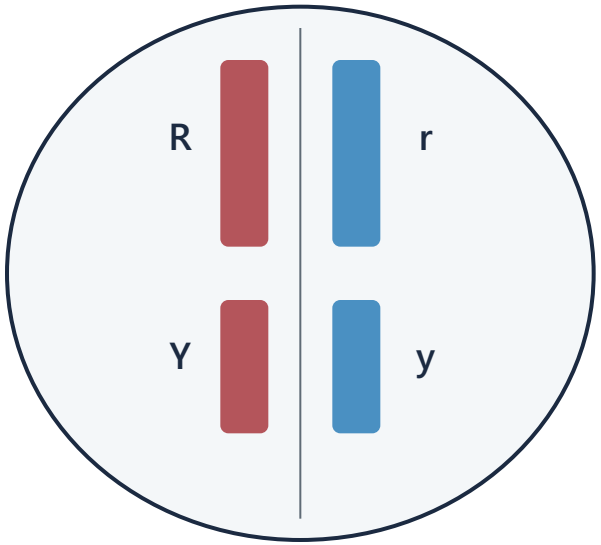


Image: peas in their pods. CC BY-SA 3.0.

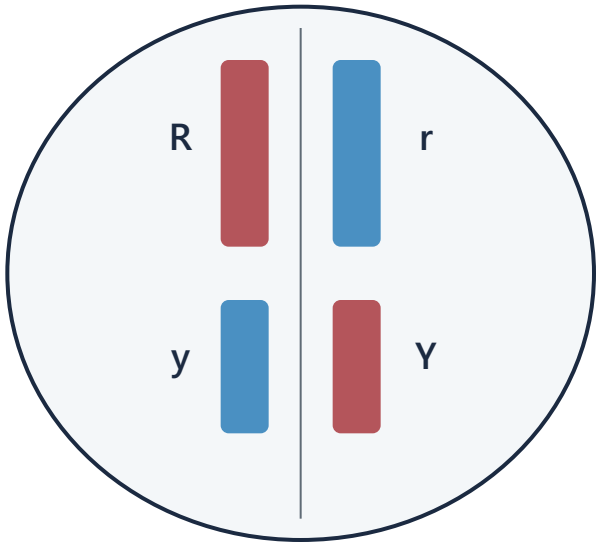
Objectives for each class

Class	Statements	By the end of the class, you can...
1	D3.2.16–17	Explain segregation and independent assortment, and use Punnett grids for dihybrid crosses
2	D3.2.18–20	Locate human genes, explain linkage, and identify recombinants in test crosses
3	D3.2.21	Use a chi-squared test on the results of a dihybrid cross

Segregation and independent assortment



gametes RY and ry



gametes Ry and rY

Segregation: the two alleles of a gene go to different gametes.

Independent assortment: pairs on different chromosomes line up at random in metaphase I, so RrYy gives **four** gamete types equally.

Only for unlinked genes

Independent assortment applies to genes on different chromosomes. Genes on the same chromosome are linked (class 2).

The 9:3:3:1 ratio

	SY	Sy	sY	sy
SY	SSYY	SSYy	SsYY	SsYy
Sy	SSYy	SSyy	SsYy	Ssyy
sY	SsYY	SsYy	ssYY	ssYy
sy	SsYy	Ssyy	ssYy	ssyy

- SsYy × SsYy (S smooth, Y yellow)
- 9 smooth yellow
- 3 smooth green
- 3 wrinkled yellow
- 1 wrinkled green

Test cross: 1:1:1:1

SsYy × ssyy gives four phenotypes equally: the ssyy parent only gives sy gametes, so each offspring shows which gamete came from the other parent.

A shortcut: multiply the probabilities

D3.2.17

Cross Ss × Ss and Yy × Yy	Smooth	Wrinkled
Yellow	3/4 × 3/4 = 9/16	1/4 × 3/4 = 3/16
Green	3/4 × 1/4 = 3/16	1/4 × 1/4 = 1/16

For unlinked genes, work out each gene separately, then **multiply**.

Quick to check a grid.

Try it

SsYy × Ssyy: what fraction of offspring are wrinkled and yellow?

Practice: dihybrid crosses

D3.2.17

1. Rabbits: brown (B) dominant to white (b); normal tail (T) dominant to no tail (t). BbTt male × bbtt female: give the gametes and the phenotype ratio.
2. Two SsTt pea plants are crossed and 320 seeds grown. How many tall plants with wrinkled seeds are expected?
3. Wwgg × wwGg in fruit flies: predict the phenotype ratio.

Check

Use a grid or the multiplication shortcut.

Working with data: Mendel’s peas

D3.2.17

F2 from a dihybrid cross of peas, as reported by Mendel (rounded).

Phenotype	Number
Round yellow	315
Round green	108
Wrinkled yellow	101
Wrinkled green	32

1. Calculate the total and the expected 9:3:3:1 numbers.
2. Do the results look close?
3. Explain why they are not exactly 9:3:3:1.

Practice: which gametes?

D3.2.16

AaBb

Unlinked genes: list the gametes and their proportions.

AaBB

List the gametes.

AaBbCc

How many kinds of gamete for three unlinked genes?

Explain

Use segregation and independent assortment in your answers.

Check-in: class 1

D3.2.16–17

1. Define segregation.
2. Explain independent assortment using meiosis.
3. State the F2 ratio from two double heterozygotes.
4. State the ratio from a test cross of AaBb × aabb.
5. Calculate the fraction aabb from AaBb × AaBb.

How to use this

Answer without your notes, then check against slides 3–8.

Where human genes are



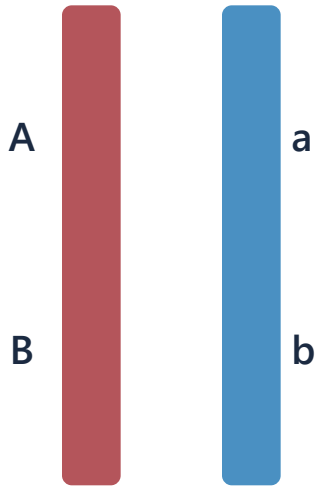
Human male karyotype. NHGRI, public domain.

Gene	Locus	Polypeptide
HBB	11p15	Beta globin
HBD	11p15	Delta globin
HBA1	16p13	Alpha globin
CFTR	7q31	Chloride channel
F8	Xq28	Clotting factor VIII

Activity

Use NCBI Gene or Ensembl to find the locus and product of three genes. Find a pair on different chromosomes and a pair close together. Which pair is likely to be inherited together?

Linked genes



AB on one chromosome, ab on the other

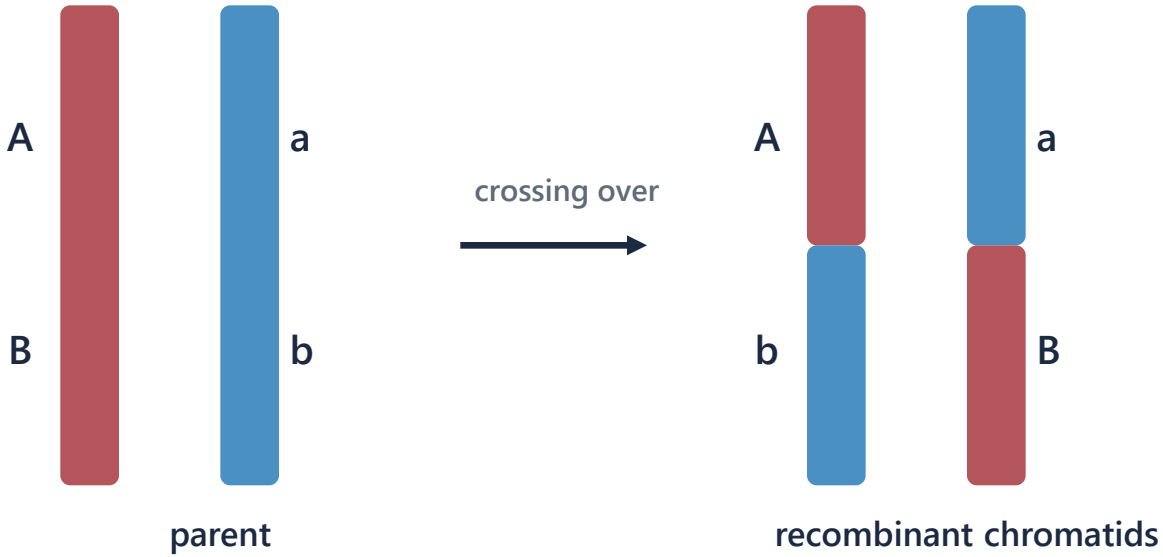
Genes on the **same chromosome** are **linked**: they tend to be inherited together and do not assort independently.

The **closer** the loci, the stronger the linkage. Alleles are written beside a vertical line for each homologue.

Why not always together?

Crossing over in prophase I can swap sections between non-sister chromatids, giving new combinations.

Recombinants



A **recombinant** has a combination of alleles not found in either parent.

In linked genes they only arise by crossing over **between** the loci, so they are fewer than parental types.

Spotting linkage in a test cross

D3.2.20

Offspring of AaBb × aabb	Unlinked genes	Linked genes
AaBb (parental)	25%	Large class
aabb (parental)	25%	Large class
Aabb (recombinant)	25%	Small class
aaBb (recombinant)	25%	Small class

Rule

Four equal classes: independent assortment. Two large and two small classes: linkage.

An example: sweet peas

D3.2.20



Sweet pea. Acabashi, CC BY-SA 4.0.

Offspring of test cross	Number	Type
Purple, long	430	Parental
White, short	420	Parental
Purple, short	75	Recombinant
White, long	75	Recombinant

Activity

Invented data. Calculate the percentage of recombinants, explain why it is far from 50%, and say what a higher percentage would mean.

Practice: linked or not?

D3.2.19–20

Result 1

A test cross gives 102, 98, 100, 100.

Result 2

A test cross gives 180, 175, 22, 23.

Result 3

Two genes 1 cm apart on chromosome 3.

For each one

Decide whether the genes are linked, and explain.

Check-in: class 2

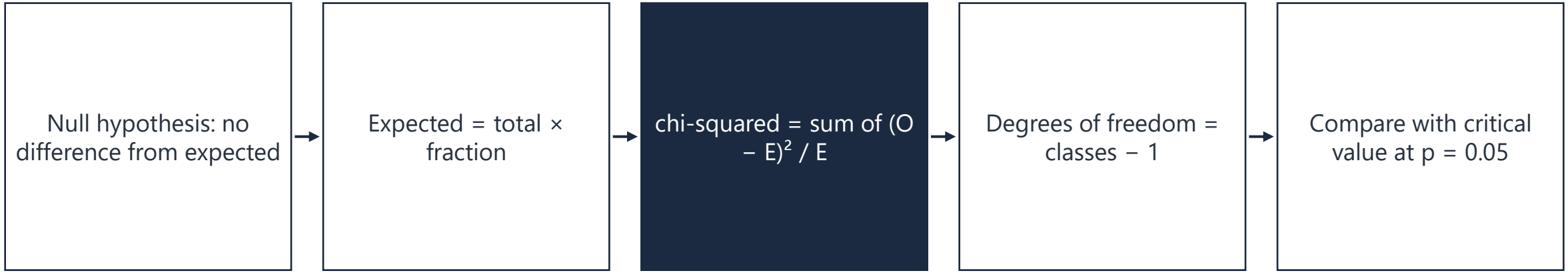
D3.2.18–20

1. Define a gene locus.
2. Define linked genes.
3. Explain why linked genes do not assort independently.
4. Explain how recombinants are produced.
5. Describe the test cross result that suggests linkage.

How to use this

Answer without your notes, then check against slides 10–15.

The chi-squared test



Decision

Calculated value **greater** than critical value: the difference is significant; reject the null hypothesis (perhaps the genes are linked). **Smaller**: the difference could be chance; do not reject. For 4 classes (3 degrees of freedom) the critical value is 7.81.

Worked example: Bateson's chickens

D3.2.21

F2 phenotype	White, pea	White, single	Dark, pea	Dark, single
Observed (O)	111	37	34	8
Expected, 9:3:3:1	106.9	35.6	35.6	11.9
$(O - E)^2 / E$	0.16	0.05	0.07	1.26

Result

Chi-squared = 1.55 with 3 degrees of freedom, below 7.81: not significant. The data fit independent assortment. (Data reported by W. Bateson, early 1900s; 190 chickens.)

Working with data: a chi-squared test

D3.2.21

A test cross AaBb × aabb gave these offspring. Counts invented for practice.

Phenotype	Observed	Expected	$(O - E)^2 / E$
AB	58	40	?
Ab	22	40	?
aB	19	40	?
ab	61	40	?

1. Complete the table and calculate chi-squared.
2. Critical value 7.81 (3 df). Conclusion?
3. Suggest an explanation.

Practice: test Mendel’s data

D3.2.21

Use Mendel’s F2 numbers from slide 7: 315, 108, 101, 32 (total 556).

1. Calculate the expected numbers for 9:3:3:1.
2. Calculate chi-squared.
3. State the degrees of freedom and compare with 7.81.
4. Conclude about independent assortment.

What chi-squared can and cannot tell you

D3.2.21

Significant

The pattern is unlikely to be chance alone: look for a cause, e.g. linkage.

Not significant

The data fit; this does not prove the hypothesis is true.

Sample size

Small samples make a real difference hard to detect.

$p = 0.05$

Means a 5% chance of a difference this big arising by chance if the null hypothesis were true.

Check-in: class 3

D3.2.21

1. State the null hypothesis for a dihybrid test cross.
2. Write the formula for chi-squared.
3. Calculate the degrees of freedom for four classes.
4. State what a value above the critical value means.
5. Explain why a non-significant result does not prove independent assortment.

How to use this

Answer without your notes, then check against slides 17–21.

Common mistakes to avoid

Not this	But this
Linked genes are on different chromosomes.	Linked genes are on the same chromosome.
A large chi-squared supports the null hypothesis.	Above the critical value: reject it.
Degrees of freedom = number of classes.	Classes minus 1.
Recombinants come from mutations.	They come from crossing over or independent assortment.

Summary: two genes at a time

D3.2.16–21

	Key idea
D3.2.16	Segregation; independent assortment in metaphase I
D3.2.17	Dihybrid 9:3:3:1; test cross 1:1:1:1
D3.2.18	Gene loci, e.g. HBB at 11p15
D3.2.19	Linked genes are on the same chromosome
D3.2.20	Recombinants from crossing over; fewer when linked
D3.2.21	Chi-squared: compare with 7.81 for 3 df

Check that you can...

1. **Explain** segregation and independent assortment
2. **Predict** dihybrid ratios using Punnett grids
3. **Outline** gene loci, linkage and recombinants
4. **Calculate** and **interpret** a chi-squared test

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the expected phenotype ratio when two double heterozygotes for unlinked genes are crossed.
2. Explain why linked genes produce few recombinants.
3. State the null hypothesis in a chi-squared test of a dihybrid cross.

Image credits

Peas in pods – Studio.jpg – Bill Ebbesen – CC BY-SA 3.0

NHGRI human male karyotype.png – Courtesy: National Human Genome Research Institute – Public domain

Sweet pea 'Lathyrus odoratus' violet flower at Boreham, Essex, England 1.jpg – Acabashi – CC BY-SA 4.0

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